Title

Spam Detection System using Machine Learning

◈ Objective

The objective of this task is to build a machine learning model that can classify text messages as **spam** or **ham (not spam)**. This helps in filtering unwanted messages automatically.

◈ Tools & Technologies Used

- Python
- Pandas
- Scikit-learn
- CountVectorizer
- Multinomial Naive Bayes

◈ Dataset Description

The dataset used in this task is a CSV file named **spam.csv**, which contains two columns:

- **text** → the message content
- **label** → category (spam or ham)

◈ Methodology

1. Data Collection

Created a dataset containing sample spam and non-spam messages.

2. Data Preprocessing

Converted all text data into lowercase to maintain consistency.

3. Feature Extraction

Used **CountVectorizer** to convert text data into numerical form.

4. Data Splitting

Split the dataset into training and testing sets using train_test_split.

5. Model Training

Trained the model using **Multinomial Naive Bayes algorithm**.

6. Model Evaluation

Evaluated the model using:

- Accuracy Score
- Classification Report

7. Prediction

Tested the model with sample messages to check whether they are spam or not.

◈ Output

- The model successfully classifies messages as spam or ham.
- Accuracy score is displayed.
- Classification report shows precision, recall, and F1-score.
- Sample predictions are generated for custom messages.

◈ Conclusion

This project demonstrates how machine learning can be used to detect spam messages effectively. It covers important steps like data preprocessing, feature extraction, model training, evaluation, and prediction. The model provides reliable results and helps in understanding real-world applications of machine learning.